XK-JS5B

Six-Freedom Electric Manipulator

Training Model



Instruction Book

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Chapter One System Overview

1.1 Product Overview

Electric manipulator is a model device for teaching. Simulate the common robot which usually used in the process of industrial production around the automation equipment.

It uses the transmission technology of friction wheel transmission、synchronization belt transmission, and also adopts the technology of stepping motor drive control、position control. It also has the function that components automatically load and unload、deliver.

This model mainly used for teaching demonstration, teaching experiment, practical training, curriculum design and scientific research, etc. It can train the students to know the mechanical structure、mechanical drive; it can train students design and debugging capacity for the hardware and software of the electrical control system; it also can train the students the analyze & solve capacity that problems happens in the actual operation process.

Electric manipulator is typical products of electromechanical integration technology. The robot's design and application with very important meaning for the engineering major of mechatronics、mechanical structure technology、automation、electronic information etc.

1.2 Technology Parameter

a. Input power: Single phase three wires～220V±10% 50Hz

b. Working environment: temperature -10℃～+40℃, relative humidity <85％ (25℃), elevation <4000m.

c. Arm rotation angle: 180°

d. Big arm rotation angle: 90°

e. Small arm rotation angle: 90°

f. Paw rotation angle: 150°

g. Finger rotation: 360°

h. Maximum load: 1KG

i. Power: ≤0.5Kw

j. Basement size: 380x200mm, total height: 950mm

k. Total net weight: 27.5KG

1.3 Product Structure

1.3.1 Whole set structure

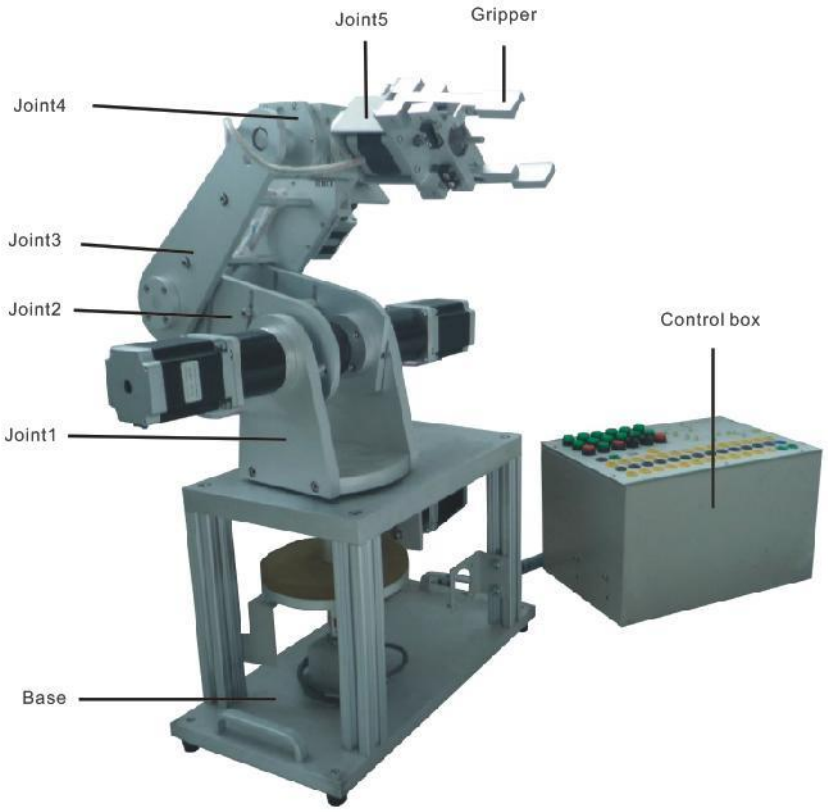


Figure 1-1

1.3.2 Unit structure

a. Foundation



Figure 1-2 Structure of foundation

Foundation is frame structure. Internal is installed with stepper motor、friction wheel pair、and rotary encoder. Friction wheel pair used for transmitting of the stepper motor torque, rotary encoder is used to record the rotation angle. Foundation part through the stepper

motor drive the friction wheel pair to supply the working torque for arm rotation in the horizontal plane.

b、Mechanical arm structure

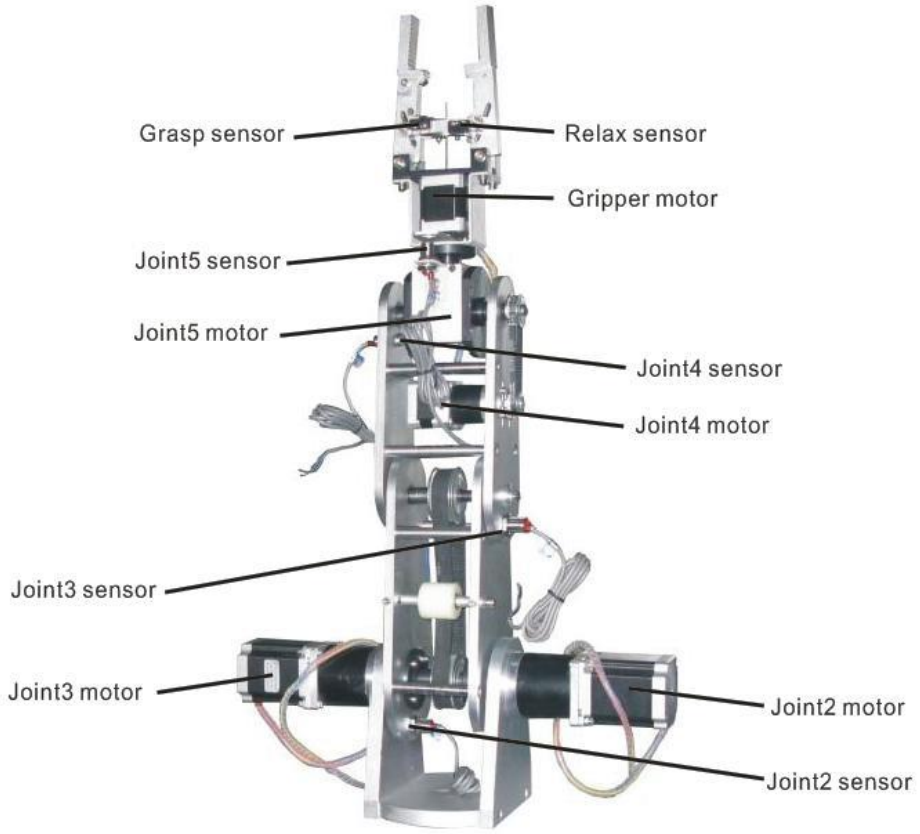


Figure 1-3 Mechanical arm structure

The mechanical arm consists of big arm、small arm and paw. The big arm mainly consists of two side plate and one keel. There is one stepper motor at the right side of big arm bottom axis, this stepper motor supplies the working torque for big arm rotation around bottom axis.

The small arm mainly consists of two side plates and two keels. There is one stepper motor at the left side of big arm bottom axis, this stepper motor through the synchronous belt drive to supply the working torque for small arm to rotation around the small arm bottom axis.

The paw mainly consists of finger、roll-over stand and stepper motor. Finger fixed on the roll-over stand. There is one stepper motor in the middle of paw, this stepper motor supplies the working torque for open and close of the two fingers.

There is one stepper motor at the tail of paw, this stepper motor and reducer supplies working torque for rotation of wrist.

There is one stepper motor at right side of small arm, this stepper motor through the synchronous drive supply working torque for whole paw rotation around the fourth axis.

c、Control panel

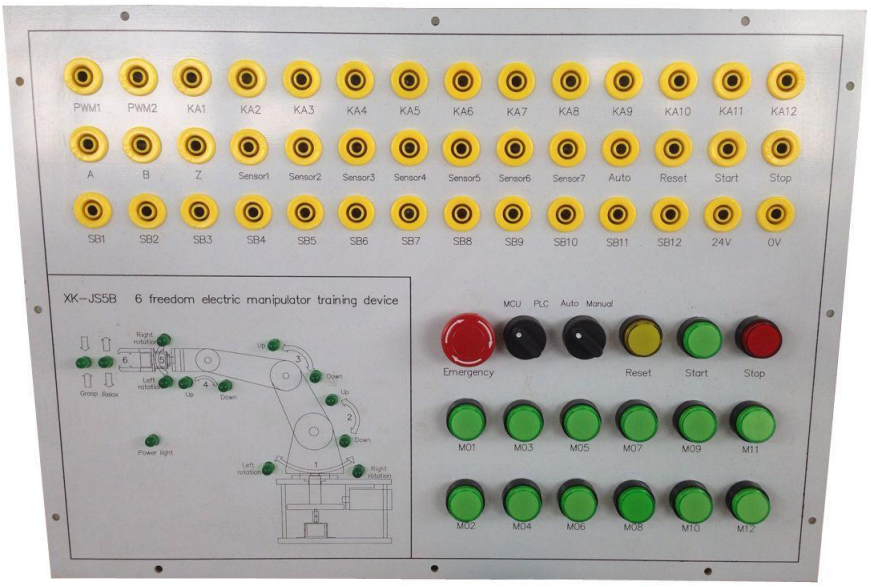


Figure 1-4 control panel

Jack definition:

PWM1: Pulse input of joint 1、2、3、4;

PWM2: Control pulse input of joint 5 and paw;

KA1: Joint 1 left turning control relay;

KA2: Joint 1 right turning control relay;

KA3: Joint 2 up control relay;

KA4: Joint 2 down control relay;

KA5: Joint 3 up control relay;

KA6: Joint 3 down control relay;

KA7: Joint 4 up control relay;

KA8: Joint 4 down control relay;

KA9: Joint 5 up control relay;

KA10: Joint 5 down control delay;

KA11: Paw firmly grasp control relay;

KA12: Paw loose control relay;

B、Z: Encoder signal input;

Sensor1-5: Joint 1-5 positioning sensor signal input;

Sensor6: Paw firmly grasp limit;

Sensor7: Paw loose limit; Auto:

Auto button input signal; Start:

start button input signal;

Stop: stop button input signal;
SB1-12: button M01-12 signal input;
Button definition:
Emergency: Emergency stop button, used for equipment emergency stop;
MCU-PLC: single chip and PLC control mode selection switch;
Auto-Manual: Auto-manual selection switch;
Start: Auto start button;
Stop: Auto stop button;
M01: Joint 1 left turning control button;
M03: Joint 1 right turning control button;
M05: Joint 2 up control button;
M07: Joint 2 down control button;
M09: Joint 3 up control button;
M11: Joint 3 down control button;
M02: Joint 4 up control button;
M04: Joint 4 down control button;
M06: Joint 5 up control button;
M08: Joint 5 down control button;
M10: Paw firmly grasp control button;
M12: Paw loose control button.

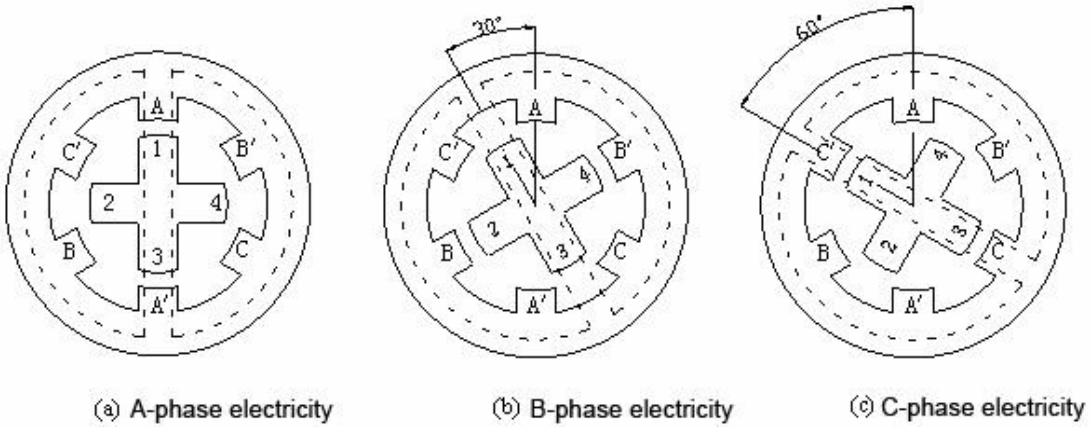
Chapter 2 Electrical control system

2.1 Stepper motor and Driver

Stepper motor is the kind of special operating motor that converts the electrical pulses signal to the corresponding angular displacement or linear displacement. For input one electrical pulse signal one time, motor will rotate one angle. Its movement type is stepper, so called the stepper motor

2.1.1 Working principle of stepper motor

The following takes one most simple three phase reaction type stepper motor as an example, introduce the working principle of stepper motor.
Figure 2-1 is principle diagram of one set three phase reaction type stepper motor.
Stator core is salient pole type, a total of three pairs (quantity of 6) magnetic pole, every two spaces on the relative polarity be around a phase control winding. The rotor made of soft magnetic materials, is also salient pole structure, there are only four teeth, tooth width is equal to the polar wide of stator.



(a) A-phase electricity (b) B-phase electricity (c) C-phase electricity

Figure 2-1

When A phase control winding get power and the rest two phase are not get electricity, in the motor, set up the magnetic field that takes the A phase polar as the axis.

Because the magnetic flux with the characteristics that pass the magnetic resistance minimum path, make the axis of the rotor teeth 1、3, and stator A phase polar axis alignment, show in Figure 2-1 (a). If A phase control winding outage、B phase control winding get power, rotor under the action of reaction torque, counterclockwise turns around 30 °, make the axis of rotator teeth 2、4 and stator B phase polar axis alignment, that's mean the rotor go one step, show in Figure 2-1 (b). if disconnect B phase, make C phase control winding get electricity, rotor anti-clockwise turn 30°, make the axis of rotor teeth 1、3 and stator C phase polar axis alignment, show in Figure 2-1 (c).

So according to the order of A—B—C—A get power take turns, rotor will be counter-clockwise direction rotation step by step. Its speed based on the frequency that each phase control winding power on or off, rotation direction based on the order that control winding power on in turn. If power on according to order of A—C—B—A, so motor will be rotation with clockwise.

The above type of get power is three phase single three clap. -three phase‖ mean three phase stepper motor; -single three clap‖ means each time only one phase to control the winding get electricity; the control winding change power on or off state one time means one clap, -three clap‖ means change the power on or off state three times as one cycle. The angle that each clap rotor turning called the stepping angle. When the three phase single three clap running, stepping angle is 30°. So, this angle is not big, cannot be real used.

If change the power on type of control winding to A→AB→B→BC→C→CA→A, namely that one phase get power then two phase get power and carry out interval in turns, finish one cycle require to change 6 times of power up state, called the three phase single、double six clap power up mode. When A、B two phase winding power on at same time, the rotor teeth position should be consider the two couple of stator pole action at same time,

only the middle position that magnetic pull is balanced which produced by A phase and B phase for rotor teeth, this middle position will be the balanced position. So, power up state that under single、double six claps, the rotor balanced position increase twice, stepping angle is 15°

Further to reduce the stepping angle is to use the stator pole with a small tooth, the rotor teeth with many structures, analysis show that this structure of the stepper motor, the interval Angle can be very small. Generally speaking, the actual stepping motor products are always use this method to realize the subdivide of the stepping angle.

2.1.2 2M542-N stepper driver

A. Performance index

2M542-N is one kind of equal angle constant torque subdivided driver, driver voltage DC 24-58V, adapter current below 4.2A, outer diameter 42,57MM, the various types of two-phase hybrid stepping motor. This driver internal adopt the circuit that similar servo control principle, this circuit can make motor running steady with low speed, almost do not have vibration and noisy, motor running is steady, heating smaller, the stepper pulse stop more than 100m/s, driver current auto to reduce half. The positioning precision can up to 25600 steps/turning

B. Main Feature

24-50V DC power supply, suitable for the worst of electric net environment

H double polar constant phase current subdivide drive

Original speed self-adaptive circuit, automatic optimization

Subdivide current setting is convenient

2-64 subdivide 15 running mode

Maximum step no. 25600Pulse/rev

Over-current protection, over-voltage protection, under-voltage protection, short circuit protection

Off-line（ENS）protection function

Work peak current range: 1.0A-4.2A, output current total 8 grades.

Current resolution ratio is 0.45A

C. Electrical index

Explanation	Min value	Typical value	Max value
Supply voltage (VDC)	24	36V	50
Normal working output current (A) peak value	1.0	-----	4.2
Logic input current (mA)	7	10	16
Stepper pulse relative level rate (KHZ)	----	----	200
Pulse low level time (US)	2.5	----	----

D. Cooling type

Cooling type	Natural cooling or forced cooling	
Using environment	Situation	Avoid dust, oil mist, and corrosive gas
	Stored temperature	-10℃~80℃
	Max. environment temperature	65℃
	Environment temperature	＜ 80%RH, non condensation without frosting
Vibration	5.9m/s²Max	
Weight	0.26kg	

E. Current setting

Dial switch			Current value （A） Average value
SW1	SW2	SW3	
1	1	1	1.00
0	1	1	1.46
1	0	1	1.91
0	0	1	2.37
1	1	0	2.84
0	1	0	3.31
1	0	0	3.76
0	0	0	4.20

F. Subdivide setting

Dial switch				Subdivide sheet
SW4	SW5	SW6	SW7	
0	1	1	1	400
1	0	1	1	800
0	0	1	1	1600
1	1	0	1	3200
0	1	0	1	6400
1	0	0	1	12800
0	0	0	1	25600
1	1	1	0	1000
0	1	1	0	2000
1	0	1	0	4000
0	0	1	0	5000
1	1	0	0	8000
0	1	0	0	10000
1	0	0	0	20000
0	0	0	0	25000

G. Input signal interface description

Mark symbol	Function	Details
PLS+	Input signal optoelectronic isolation plus end	Connect +5V power supply, +5V~+24V all can drive, if higher than +5V required connect current-limiting resistance
PLS-	Stepper pulse signal	Falling edge effective, each time pulse from high to low, the motor goes further one step. Input resistance 220Ω , requirement: low level 0-0.5V, high level 4-5V, pulse width >2.5uS
DIR+	Input signal optoelectronic isolation plus end	Connect +5V supply power, +5V~+24V all can Be driven, higher than +5V required connect current –limiting resistance
DIR-	Direction control signal	Used for change direction of motor rotation, input resistance 220Ω , requirement: low level 0-0.5V, high level 4-5V, pulse width >2.5uS
ENA+	Input signal optoelectronic isolation plus end	Connect +5V supply power, +5V-+24V all can drive, higher than +5V require to connect current-limiting resistance
ENA-	Motor release signal	Effective (low level) turn off motor coil current, drive stop working, motor under the free state.

H. Motor wiring

The use of stepper motor, one is to pay attention the correct installation, the second is the correct wiring.

Installation of the stepper motor, It must be in strict accordance with the requirements of the product description. Stepper motor is one kind of precision device, pay attention that, it's not allow to beat its shaft end during installation and even don't remove the motor.

Wiring is different for different stepper motor, the 4 wire stepper motor wiring diagram show as Figure 2-2, six wires system wiring diagram show in Figure 2-3 (a)

There are two kinds of wiring for 8 wires system: parallel connection and series connection, parallel wiring is high speed constant power connection method, show in Figure 2-3 (b), series wiring is high torque low speed connection method, show in Figure 2-3 （c）.

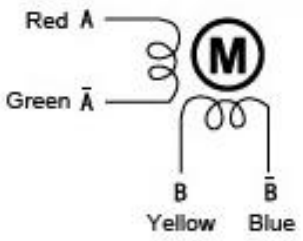


Figure 2-2

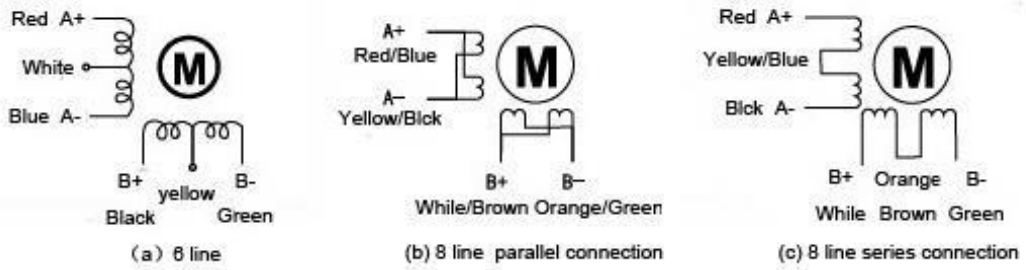


Figure 2-3

2.1.3 Note for stepper motor using

During control the running of stepper motor, it should pay attention that consider about prevent the step out of stepping motor running.

Stepper motor step out include lose step and over step. During lose step, the number of rotor forward steps is smaller than the pulse number, during the over step, the number of rotor forward steps is more than pulse number. When lose step is serious, will make the rotor stay in one place or vibration around one position; when the over step is serious, the device will be happen overshoot.

The operations that make manipulator return to the origin, usually happen the situation that over step. When the manipulator device back to the origin, the origin switch action, make the command input OFF. But the speed is too high before reaching the origin. The inertia torque will be bigger than the holding torque of stepping motor and make the

stepper motor over step. So the operation that go back to the origin should ensure enough low-speed advisable; When the stepper motor to drive the manipulator running at a high speed, it emergency stop, it's unavoidable to happen over step, therefore it should be first return to the origin with low speed, then restore the original method of the operation.

Due to the motor winding itself is the inductive load, the higher the input frequency, the less the exciting current. High frequency, magnetic flux change greatly, eddy current losses become large. Therefore, higher of input frequency, output torque is reduced. Output torque of maximum working frequency can just achieve 40-50% of low frequency torque. During control of the high-speed positioning, if the specified frequency is too high, it will be happen the lose step phenomenon.

Output torque of maximum working frequency can just achieve 40-50% of low frequency torque. During control of the high-speed positioning, if the specified frequency is too high, it will be happen the lose step phenomenon.

In addition, if the mechanical parts cannot adjust well, it can make the mechanical load increase. Stepper motor can't be load running, even a moment, it will happen out of step, When serious situation it will be halt or vibration origin irregularly over and over again.

2.2 Operation

2.2.1 System operation and demo

There are two types of running mode: manual running and auto running. Manual running can control each joint running of the manipulator. Auto running can realize the auto carrying running of manipulator. User can program for the PLC, according to the user requirement to finish different motion control.

First, it should connect the manipulator and control box with the aviation connector and connection wires; then connect one end of power line to the control box、the other end is inserted into AC220V power socket. Open the power main switch at the right side of control box, clockwise rotation of the emergency switch, the emergency switch is upspring, the power indicator is light on. To do manual or auto demo, first turn the -MCU-PLC‖ selection switch to left side -MCU‖.

Manual running operation:

Turn the -Auto-Manual‖ selection switch to right side of -Manual‖, press -M01‖ button, now the manipulator do the left rotation running, loose the button and stop running;

Press -M02‖ button, now manipulator do right rotation, loose button stop running;

Press -M03‖ button, now the manipulator joint 2 is go up, loose the button stop running;

Press -M04‖ button, now the manipulator joint 2 is go down, loose the button stop running;

Press -M05‖ button, now the manipulator joint 3 is go up, loose the button stop running;
Press -M06‖ button, now the manipulator joint 3 is go down, loose the button stop running;
Press -M07‖ button, now the manipulator joint 4 is go up, loose the button stop running;
Press -M08‖ button, now the manipulator joint 4 is go down;
Press -M09‖ button, now the manipulator joint 5 left rotation, loose the button, the manipulator stop running;
Press -M10‖ button, now the manipulator joint 5 right rotation, loose the button, the manipulator stop running;
Press -M11‖ button, now the manipulator paw is firmly grasp, loose the button and stop running;
Press -M12‖ button, now the manipulator paw is release.

Auto running operation: before the auto running, first should adjust the joint 1 to original position by manual, press -M01‖ or -M02‖ button, the manipulator runs to the original point position and stop automatically.
Then turn the selection switch -Auto-Manual‖ to left side of -Auto‖ side, then all other joints of manipulator starts to find original point position automatically, after finish, press button -start‖, the system start auto running, the process is: carry the object from start position to the designated spot, put down the object, and come back automatically, start the next carrying, if press button -stop‖ before it automatically come back to the original position, the system will stop as soon as it come back to the original point position.

2.2.2 Preparation and inspection before using
a. Be sure the power reliable ground connection;
b. Check that the friction wheel set is reliable mesh;
c. Check that synchronous belt is reliable connection;
d. Check that the connection between the stepper motor and Driver is right.
e. Check that connection between each Driver and controller is right.
f. Check that is there something in the paw.
g. Check that all the joints should be in the range of the limit switch.

2.2.3 Fault analysis and trouble shooting

Fault phenomenon	Reason	Trouble shooting
Arm cannot rotation round the foundation	1、 Friction wheel set did not mesh well 2、 Driver and stepper motor without reliable connection 3、 Controller and driver did not reliable connection	1、 Adjusting mesh size of friction wheel set 2、 To re-connect the Driver and stepper motor 3、 Re-connect the controller and Driver
Rotary encoder do not play action	1、 Rotary encoder is bad 2、 Did not connect well with shaft	1、 Change the new rotary encoder 2、 Adjust the connection with shaft
The whole paw cannot rotation	1、 The distance between the two synchronous pulley is inappropriate. 2、 The Driver and stepper motor did not connect reliable 3、 The controller and Driver did not connect reliable	1、 Adjust the wheel distance of synchronous pulley 2、 To re-connect the Driver and stepper motor. 3、 To re-connect the controller and Driver
Small arm cannot rotation	1、The distance between the two synchronous pulley is inappropriate. 2、There is no reliable connection between the Driver and stepper motor 3、There is no reliable connection between the controller and Driver 4、Synchronous belt badly worn	1、Adjust the wheel distance of synchronous pulley 2、To re-connect the Driver and stepper motor 3、To re-connect the controller and Driver 4、Change the synchronous belt
Finger cannot open and close	1、 There is no reliable connection between the controller and Driver 2、 There is no reliable connection between the Driver and stepper motor	1、 To re-connect the controller and Driver 2、 To re-connect the Driver and stepper motor
Wrist cannot rotation	1、 Shaft connection is abnormal 2、 There is no reliable connection between the Driver and stepper motor 3、 There is no reliable connection between the controller and Driver	1、Adjust the shaft connection 2、To re-connect the Driver and stepper motor 3、To re-connect the controller and Driver
Big arm cannot rotation	1、There is no reliable connection between the Driver and stepper motor 2、There is no reliable connection between the controller and Driver	1、To re-connect the Driver and stepper motor 2、To re-connect the controller and Driver

Attachment A Terminals strip wiring diagram

1	Motor1-A+		26	Sensor1-0V
2	Motor1-A-		27	Sensor1-output
3	Motor1-B+		28	Sensor2-24V
4	Motor1-B-		29	Sensor2-0V
5	Motor2-A+		30	Sensor2-output
6	Motor2-A-		31	Sensor3-24V
7	Motor2-B+		32	Sensor3-0V
8	Motor2-B-		33	Sensor3-output
9	Motor3-A+		34	Sensor4-24V
10	Motor3-A-		35	Sensor4-0V
11	Motor3-B+		36	Sensor4-output
12	Motor3-B-		37	Sensor5-24V
13	Motor4-A+		38	Sensor5-0V
14	Motor4-A-		39	Sensor5-output
15	Motor4-B+		40	Grasp limit
16	Motor4-B-		41	Limit com
17	Motor5-A+		42	Relax limit
18	Motor5-A-		43	Encoder-24V
19	Motor5-B+		44	Encoder-0V
20	Motor5-B-		45	Encoder-A
21	Motor6-A+		46	Encoder-B
22	Motor6-A-		47	Encoder-Z
23	Motor6-B+		48	
24	Motor6-B-		49	
25	Sensor1-24V		50	